Assalamualaikum Sir,

1. You advised me to learn ACO and GA from the YouTube channel The Ali Mirjalili. I have gone through the videos, but GA is taught by him in a paid Udemy course, which I am unable to access.
2. You also instructed me to read the paper *Optimal Path Planning Based on ACO in Intelligent Transportation*. After carefully reviewing the paper multiple times, I found that the authors do not implement real-world traffic data directly. Instead, they model the problem in relation to Vehicle Routing Problems (VRP) and Vehicle Control Problems (VCP)**.**

   Their approach combines Genetic Algorithm (GA**)** and Ant Colony Optimization (ACO) with a reward-and-punishment mechanism. In their framework, a route with a lower pheromone level is interpreted as a congested path. To manage this, they introduce a reward system for selecting alternative, less congested routes and apply punishment for choosing highly congested ones.

In summary, the paper focuses on a simulation-based congestion management strategy rather than actual real-time traffic data, using GA–ACO hybrid optimization linked with VRP/VCP formulations.

Other papers links and related formulas in this paper are:

1.

Paper: on (vcp)

Konstantakopoulos, G.D., Gayialis, S.P. & Kechagias, E.P. Vehicle routing problem and related algorithms for logistics distribution: a literature review and classification. *Oper Res Int J* **22**, 2033–2062 (2022).

Link:

https://doi.org/10.1007/s12351-020-00600-7

this is the main task you assigned me to find .

2. Reward–punishment fitness adjustment (related)

Paper:

Qingqing Sun, Hong Chen, Ruyin Long, Jiahui Yang,

Who will pay for the “bicycle cemetery”? Evolutionary game analysis of recycling abandoned shared bicycles under dynamic reward and punishment,

European Journal of Operational Research,

Volume 305, Issue 2,

2023,

Pages 917-929,

ISSN 0377-2217,

Link:

https://doi.org/10.1016/j.ejor.2022.06.013

3.Initial pheromone setting in RA-GA-ACO

Paper:

Mazaheri, H., Goli, S. & Nourollah, A. Path planning in three-dimensional space based on butterfly optimization algorithm. *Sci Rep* **14**, 2332 (2024).

Link:

https://doi.org/10.1038/s41598-024-52750-9

4.

Paper:

Yang, Liwei & Fu, Lixia & Li, Ping & Mao, Jianlin & Guo, Ning & Du, Linghao. (2021). LF-ACO: an effective formation path planning for multi-mobile robot. Mathematical Biosciences and Engineering. 19. 225-252. 10.3934/mbe.2022012.

Link:

https://www.aimspress.com/article/doi/10.3934/mbe.2022012

5. Pheromone diffusion under reward–punishment

This looks like a new contribution of this paper (no external citation).

6.

Paper:

H. Inahara and N. Motoi, “Research on search algorithm using particle swarm optimization with virtual pheromone for swarm robots,” *IEEJ Journal of Industry Applications*, vol. 11, no. 4, pp. 634–642, 2022,

Link:

https://doi.org/10.1541/ieejjia.21000994